ForgeMind — Continual Learning Operational AI

Factories run on undocumented tribal knowledge. ForgeMind is an AI system built to capture, verify, govern, and continuously improve that knowledge — without self-poisoning.

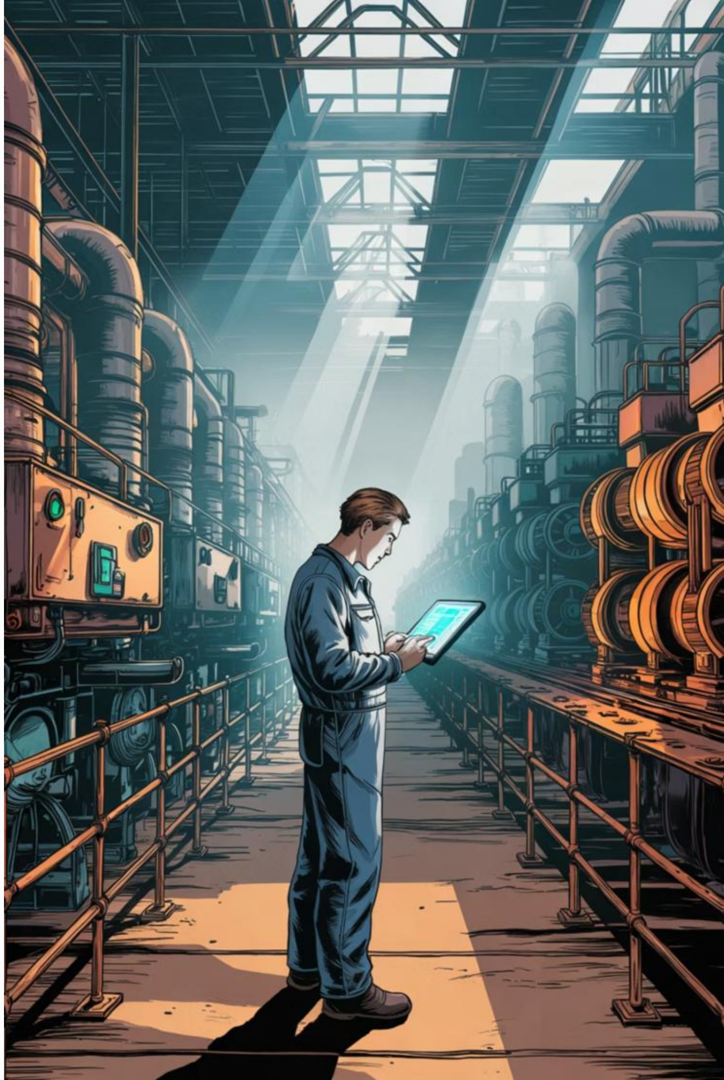
The Problem

- Machine behavior lives in operators' heads
- Heuristics and unofficial fixes are never written down
- Traditional AI only reads SOPs, manuals, and static docs

The Goal

- Learn from workers in real time
- Verify and govern operational claims
- Detect conflicts, prevent self-poisoning
- Continuously improve operational memory

The hard problem is not extraction. The hard problem is **operational knowledge governance**.



Architecture Decisions

Three foundational design choices shape ForgeMind's reliability and deployability.

Structured Memory

Knowledge is stored as **entity → condition → action → outcome** tuples — not raw chat history. This makes verification tractable, conflict detection possible, and governance workflows feasible.

Human Review Queue

Workers can be wrong, joking, or out of date. Suspicious claims are **quarantined** before entering memory. Supervisors approve or reject before anything becomes operational truth.

Local LLM + FastAPI

Built on **Ollama + TinyLlama + FastAPI** for low-cost inference, edge deployability, and privacy-preserving operation — no cloud dependency required.

Preventing Knowledge Poisoning

Blind continual learning is dangerous. Contradictory worker inputs must be caught before they corrupt operational memory.

The Conflict Problem

Worker A says: *"Reduce current by 5%."*
Worker B says: *"Increase current by 5%."*
A naive system accepts both. ForgeMind doesn't.

Human Governance

Once a conflict is resolved, approved claims **replace outdated memory** and prevent contradictory retrieval in future interactions.

⚠ Human approval alone is insufficient. Operational memory must also **reconcile contradictory truths** — not just accept the latest input.

Detect

Flag contradictions at ingestion

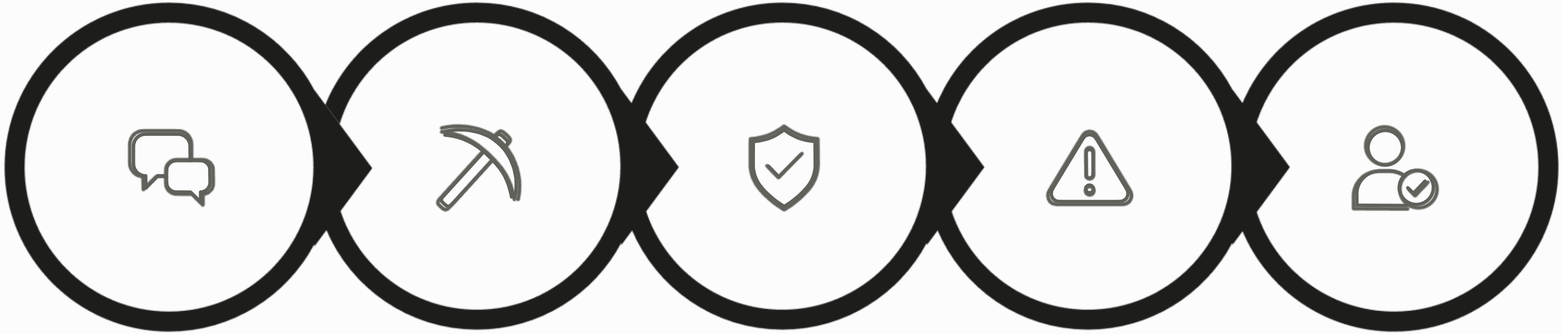
Quarantine

Hold suspicious claims out of memory

Escalate

Route to human supervisor for resolution

Continual Operational Learning Flow



Conversation

Extraction

Verification

Conflict

Human Review

Measurable Learning Signals

The system tracks accepted claims, active conflicts, quarantined inputs, and retrieval hits — providing a live audit trail of operational knowledge health.

Temporal Memory

Every claim is timestamped. Outdated knowledge is filtered out and recent operational behavior is prioritized, ensuring memory reflects current factory reality.

Limitations & Future Improvements

Current Prototype Constraints

- Partly rule-based extraction — limited semantic depth
- Simple retrieval filtering — no vector search yet
- No real-time speech streaming support

Planned Upgrades

- Embedding-based semantic memory + vector retrieval
- GraphRAG for relational knowledge traversal
- Adaptive worker trust scoring
- Speech-to-text pipeline + contextual operational reasoning

ForgeMind's focus is **operational knowledge governance** — not just chatbot interaction. The architecture is designed to scale toward production-grade industrial deployment.